Plane

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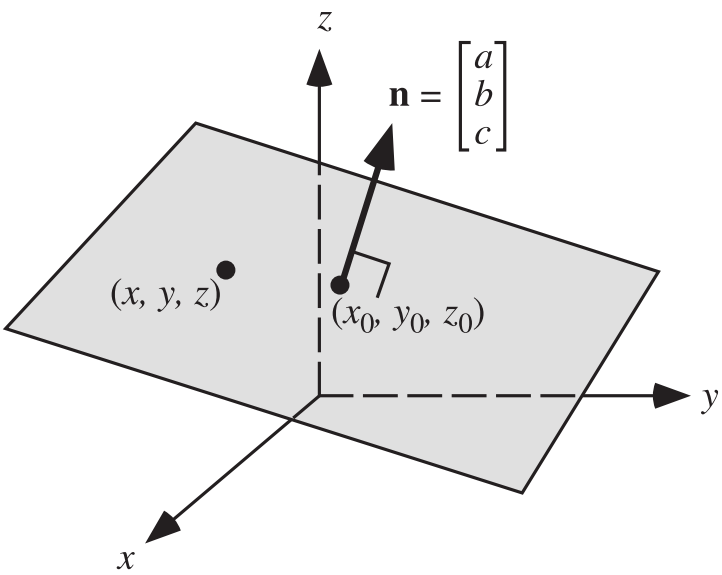
A plane is a two-dimensional doubly ruled surface spanned by two linearly independent vectors. The generalization of the plane to higher dimensions is called a hyperplane. The angle between two intersecting planes is known as the dihedral angle.



The equation of a plane with nonzero normal vector $\mathbf{n} = (a, b, c)$ through the point $\mathbf{x}_0 = (x_0, y_0, z_0)$ is

$$\mathbf{n} \cdot (\mathbf{x} - \mathbf{x}_0) = 0, \tag{1}$$

where $\mathbf{x} = (x, y, z)$. Plugging in gives the general equation of a plane,

$$a\,x + b\,y + c\,z + d = 0, \tag{2}$$

where

$$d \equiv -a\,x_0 - b\,y_0 - c\,z_0. \tag{3}$$

A plane specified in this form therefore has x-, y-, and z-intercepts at

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$$y = -\frac{d}{b} \tag{5}$$

$$z = -\frac{d}{c}, \tag{6}$$

and lies at a distance

$$D = \frac{d}{\sqrt{a^2 + b^2 + c^2}} \tag{7}$$

from the origin.

It is especially convenient to specify planes in so-called Hessian normal form. This is obtained from (◇) by defining the components of the *unit* normal vector $\hat{\mathbf{n}} = (n_x, n_y, n_z)$

$$n_x = \frac{a}{\sqrt{a^2 + b^2 + c^2}} \tag{8}$$

$$n_y = \frac{b}{\sqrt{a^2 + b^2 + c^2}} \tag{9}$$

$$n_z = \frac{c}{\sqrt{a^2 + b^2 + c^2}} \tag{10}$$

and the constant

$$p = \frac{d}{\sqrt{a^2 + b^2 + c^2}}. \tag{11}$$

Then the Hessian normal form of the plane is

$$\hat{\mathbf{n}} \cdot \mathbf{x} = -p \tag{12}$$

(Gellert *et al.* 1989, p. 540), the (signed) distance to a point $\mathbf{x}_0$ is

(13)

and the distance from the origin is simply

$$D = p \tag{14}$$

(Gellert *et al.* 1989, p. 541).

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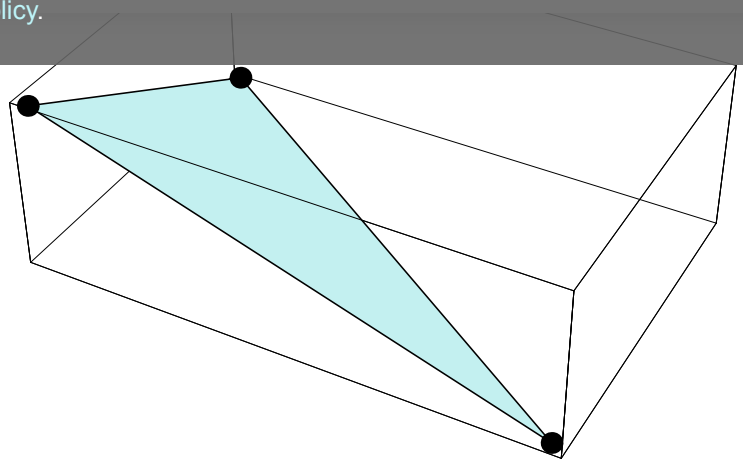

In intercept form, a plane passing through the points , and is given by

(15)

The plane through and parallel to and is

$$\begin{vmatrix} x-x_1 & y-y_1 & z-z_1 \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix} = 0. \tag{16}$$

The plane through points and parallel to direction is

$$\begin{vmatrix} x-x_1 & y-y_1 & z-z_1 \\ x_2-x_1 & y_2-y_1 & z_2-z_1 \\ a & b & c \end{vmatrix} = 0. \tag{17}$$

The three-point form is

(18)

A plane specified in three-point form can be given in terms of the general equation (◇) by

(19)

where

(20)

and is the determinant obtained by replacing with a column vector of 1s. To express in Hessian normal form, note that the unit normal vector can also be immediately written as

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and the constant giving the distance from the plane to the origin is

(22)

The (signed) point-plane distance from a point to a plane

(23)

is

(24)

The dihedral angle between the planes

(25)

(26)

which have normal vectors and is simply given via the dot product of the normals,

(27)

(28)

The dihedral angle is therefore particularly simple to compute if the planes are specified in Hessian normal form (Gellert *et al.* 1989, p. 541).

In order to specify the relative distances of points in the plane, coordinates are needed, since the first can always be placed at (0, 0) and the second at , where it defines the *x*-axis. The remaining points need two coordinates each. However, the total number of distances is

(29)

where is a binomial coefficient, so the distances between points are subject to relationships, where

(30)

For and , there are no relationships. However, for a quadrilateral (with), there is one (Weinberg 1972).

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intersect in exactly one point. For every set of points in the plane, there exists a point in the plane having the property such that *every* straight line through has at least 1/3 of the points on each side of it (Honsberger 1985).

Every rigid motion of the plane is one of the following types (Singer 1995):

1. Rotation about a fixed point .

2. Translation in the direction of a line .

3. Reflection across a line .

4. Glide-reflections along a line .

Every rigid motion of the hyperbolic plane is one of the previous types or a

5. Horocycle rotation.

SEE ALSO

Argand Plane, Complex Plane, Cox's Theorem, Cross Section, Dihedral Angle, Director, Doubly Ruled Surface, Elliptic Plane, Fano Plane, Hessian Normal Form, Hyperplane, Isoclinal Plane, Line-Plane Intersection, Mediator, Moufang Plane, Nirenberg's Conjecture, Normal Section, Plane at Infinity, Plane-Plane Intersection, Point-Plane Distance, Projective Plane

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